SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE		DEPARTMENT OF COMPUTER SCIENCE ENGINEERING	
ProgramName: B. Tech		**Assignment Type: Lab**	**AcademicYear:** 2025-2026
CourseCoordinatorName		Venkataramana Veeramsetty	
Instructor(s)Name		Dr. V. Venkataramana (Co-ordinator) Dr. T. Sampath Kumar Dr. Pramoda Patro Dr. Brij Kishor Tiwari Dr.J.Ravichander Dr. Mohammand Ali Shaik Dr. Anirodh Kumar Mr. S.Naresh Kumar Dr. RAJESH VELPULA Mr. Kundhan Kumar Ms. Ch.Rajitha Mr. M Prakash Mr. B.Raju Intern 1 (Dharma teja) Intern 2 (Sai Prasad) Intern 3 (Sowmya) NS_2 (Mounika)	
CourseCode	24CS002PC215	**CourseTitle**	AI Assisted Coding
Year/Sem	II/I	**Regulation**	R24
Date and Day of Assignment	Week4 - Tuesday	**Time(s)**	
Duration	2 Hours	**Applicableto Batches**	
AssignmentNumber:7.2(Present assignment number)/**24**(Total number of assignments)			

Q.No.	Question	*ExpectedTime to complete*
1	Lab 7: Error Debugging with AI: Systematic approaches to finding and fixing bugs **Lab Objectives:** • To identify and correct syntax, logic, and runtime errors in Python programs using AI tools. • To understand common programming bugs and AI-assisted debugging suggestions.	Week4 - Wednesday

- To evaluate how AI explains, detects, and fixes different types of coding errors.
- To build confidence in using AI to perform structured debugging practices.

Lab Outcomes (LOs):
After completing this lab, students will be able to:

- Use AI tools to detect and correct syntax, logic, and runtime errors.
- Interpret AI-suggested bug fixes and explanations.
- Apply systematic debugging strategies supported by AI-generated insights.
- Refactor buggy code using responsible and reliable programming patterns.

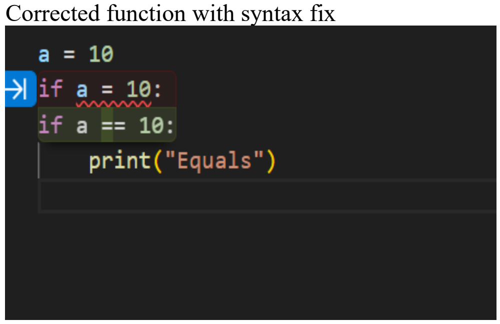
Task Description#1

- Task #1 – Syntax Error in Conditionals

```python
a = 10
if a = 10:
    print("Equal")
```

Expected Output#1

- Corrected function with syntax fix



Task Description#2 (Loops)

- Task #2 – Loop Off-By-One Error.

```python
def sum_upto_n(n):
    total = 0
    for i in range(1, n):
        total += i
    return total
```

Expected Output#2

- AI fixes increment/decrement error

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```python
def sum_upto_n(n):
    total = 0
    for i in range(1, n + 1):
        total += i
    return total
```

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Task Description#3

- Error: AttributeError

```python
class User:
    def __init__(self, name):
        self.name = name

u = User("Alice")
print(u.getName())
```

Expected Output#3

- Identify the missing method and correct the code.

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```python
class user:
    def __init__(self, name):
        self.name = name
    def getname(self):
        return self.name
u=user("Alice")
print(u.getname())
```

Task Description#4

- **Incorrect Class Attribute Initialization**

```python
class Car:
    def start():
        print("Car started")

mycar = Car()
mycar.start()
```

Expected Output#4

- Detect missing self and initialize attributes properly.

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```python
class Car:
    def start(self):
        print("Car started")
    mycar = Car()
    mycar.start()
```

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Task Description#5

- Conditional Logic Error in Grading System

```python
def grade_student(score):
    if score < 40:
        return "A"
    elif score < 70:
        return "B"
    else:
        return "C"
```

Expected Output#5

- Detect illogical grading and correct the grade levels.

This is a logical error so we fix the code by asking copilot.

Here's the corrected logic:

```python
def grade_student(score):
    if score >= 70:
        return "A"
    elif score >= 40:
        return "B"
    else:
        return "C"
```

Note: Report should be submitted a word document for all tasks in a single document with prompts, comments & code explanation, and output and if required, screenshots

Evaluation Criteria:

Criteria	Max Marks
Task #1	0.5
Task #2	0.5
Task #3	0.5
Task #4	0.5
Task #5	0.5
Total	**2.5 Marks**